

Desflurane anesthesia after sevoflurane inhaled induction reduces severity of emergence agitation in children undergoing minor ear-nose-throat surgery compared with sevoflurane induction and maintenance.

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Emergence agitation may occur after general anesthesia with volatile anesthetics in children. We designed this study to examine the emergence behavior of children undergoing ear-nose-throat surgery after sevoflurane induction and desflurane maintenance versus both sevoflurane induction and maintenance using a recently published Pediatric Anesthesia Emergence Delirium (PAED) scale. In 38 premedicated children aged 12 mo to 7 yr mask induction with sevoflurane was performed and they were randomly assigned to receive either sevoflurane ($n = 19$) or desflurane ($n = 19$) for maintenance of general anesthesia. Time to tracheal extubation, modified Aldrete score, emergence behavior, recovery complications, and pain scores were assessed. The PAED scale showed a significant advantage for desflurane (6 [0-15] versus 12 [2-20], maximum total score of 20 for severe agitation). Time to extubation was significantly shorter with desflurane than with sevoflurane (5.4 $\pm$ 1.4 versus 13.4 $\pm$ 1.8 min). The modified Aldrete score on arrival in the postanesthesia care unit (PACU) was significantly lower in children receiving sevoflurane for maintenance. Time to discharge from PACU to normal ward and the incidence of adverse effects were not significantly different between the groups. In conclusion, the use of desflurane for maintenance of anesthesia after sevoflurane induction in children is associated with less severe emergence agitation and faster emergence times.

Comparing incidence of emergence delirium between sevoflurane and desflurane in children following routine otolaryngology procedures.

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Emergence delirium (ED) is a state of aggressive agitation that can occur temporarily in the process of emerging from anesthesia in children exposed to volatile or intravenous anesthetics. Emergence delirium is typically assessed using the published and validated Pediatric Emergence Delirium (PAED) Scale. Due to some variation in properties between sevoflurane and desflurane for maintenance of anesthesia after standard sevoflurane induction, we designed a prospective study to examine potential differences in emergence behavior and incidence of ED in children undergoing elective ear-nose-throat surgery. Forty-six children aged 12 months-7 years were randomly assigned to receive either sevoflurane ($N=23$) or desflurane ($N=23$) for maintenance of general anesthesia. All patients were extubated awake in the OR, and upon arrival in the PACU, PAED scores were assessed every 15 minutes until discharged. In addition to PAED scores, time to tracheal extubation, emergence behavior, pain scores, and recovery complications were recorded. We found no significant difference in incidence of ED or peak PAED scores between sevoflurane and desflurane groups (12 [0-18] versus 12 [0-20]; $P=0.79$). There were no significant differences between desflurane and sevoflurane with respect to incidence of adverse events, such as nausea, vomiting, laryngospasm, or excessive secretions. In conclusion, the use of desflurane for maintenance of anesthesia did not significantly affect the incidence or duration of ED when compared to sevoflurane. However, desflurane did not demonstrate any increase in adverse events, which may support its routine use in this patient population.

Volatile anesthetics induce caspase-dependent, mitochondria-mediated apoptosis in human T lymphocytes in vitro.

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Volatile anesthetics modulate lymphocyte function during surgery, and this compromises postoperative immune competence. The current work was undertaken to examine whether volatile anesthetics induce apoptosis in human T lymphocytes and what apoptotic signaling pathway might be used. Effects of sevoflurane, isoflurane, and desflurane were studied in primary human CD3 T lymphocytes and Jurkat T cells in vitro. Apoptosis and mitochondrial membrane potential were assessed using flow cytometry after green fluorescent protein-annexin V and DiOC6-fluorochrome staining. Activity and proteolytic processing of caspase 3 was measured by cleaving of the fluorogenic effector caspase substrate Ac-DEVD-AMC and by anti-caspase-3 Western blotting. Release of mitochondrial cytochrome c was studied after cell fractionation using anti-cytochrome c Western blotting and enzyme-linked immunosorbent assays. Sevoflurane and isoflurane induced apoptosis in human T lymphocytes in a dose-dependent manner. By contrast, desflurane did not exert any proapoptotic effects. The apoptotic signaling pathway used by sevoflurane involved disruption of the mitochondrial membrane potential and release of cytochrome c from mitochondria to the cytosol. In addition, the authors observed a proteolytic cleavage of the inactive p32 procaspase 3 to the active p17 fragment, increased caspase-3-like activity, and cleavage of the caspase-3 substrate poly-ADP-ribose-polymerase. Sevoflurane-induced apoptosis was blocked by the general caspase inhibitor Z-VAD.fmk. Death signaling was not mediated via the Fas/CD95 receptor pathway because neither anti-Fas/CD95 receptor antagonism nor FADD deficiency or caspase-8 deficiency were able to attenuate sevoflurane-mediated apoptosis. Sevoflurane and isoflurane induce apoptosis in T lymphocytes via increased mitochondrial membrane permeability and caspase-3 activation, but independently of death receptor signaling.

Pharmacology of sedative-analgesic agents: dexmedetomidine, remifentanil, ketamine, volatile anesthetics, and the role of peripheral mu antagonists.

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In this article, the authors discuss the pharmacology of sedative-analgesic agents like dexmedetomidine, remifentanil, ketamine, and volatile anesthetics. Dexmedetomidine is a highly selective alpha-2 agonist that provides anxiolysis and cooperative sedation without respiratory depression. It has organ protective effects against ischemic and hypoxic injury, including cardioprotection, neuroprotection, and renoprotection. Remifentanil is an ultra-short-acting opioid that acts as a mu-receptor agonist. Ketamine is a nonbarbiturate phencyclidine derivative and provides analgesia and apparent anesthesia with relative hemodynamic stability. Volatile anesthetics such as isoflurane, sevoflurane, and desflurane are in daily use in the operating room in the delivery of general anesthesia. A major advantage of these halogenated ethers is their quick onset, quick offset, and ease of titration in rendering the patient unconscious, immobile, and amnestic.

What is new with inhaled anesthetics: Part 1.

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Inhaled anesthetic agents are widely used in the maintenance of general anesthesia. The newer agents, desflurane and sevoflurane, possess pharmacokinetic profiles that are distinctly different from the older inhaled anesthetics. These differences significantly impact the PACU nursing care. This article will focus on the uptake, distribution, and elimination of inhaled anesthetic agents with emphasis being placed on the postanesthesia care of surgical patients receiving these newer agents.

[Application of desflurane in anaesthesiology].

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Desflurane (Des)--is a modern inhalation anaesthetic available in Russia since August, 2013. Des is a halogenated ether; its chemical structure is 2-difluoromethoxy-1-1-1-2-tetrafluoroethane (C₃H₂F₆O). Special thermocompensated evaporators are used for Des dosing. Low solubility in blood and tissues of an organism causes fast absorption and elimination of Des. Blood/gas distribution ratio of Des is 0.42. Des distinctive properties are high saturated vapor pressure, super short duration of action and average power. Furthermore it is characterized by the minimal metabolism and lack of interaction with soda lime. Des is used for general anesthesia in a cardiac surgery neurosurgery, out-patient surgery, pediatric practice and other areas of surgery. Des has more positive qualities and fewer limitations, than other inhalation anaesthetics (halothane, isoflurane, sevoflurane). High cost of the anaesthetic is compensated by quality and controllability of anaesthesia and reduction of stay time in recovery unit. Fast elimination of the anaesthetic from a body allows reducing a frequency of complications connected with violation of upper airway and hypoxemia, promotes early discontinuation of artificial ventilation, reducing somnolence, earlier restoring a muscular tone in the postoperative period.

Sevoflurane concentrations in blood, brain, and lung after sevoflurane-induced death.

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Sevoflurane concentrations in blood, brain, and lung were measured in an individual apparently dying from sevoflurane inhalation. Sevoflurane is a volatile nonflammable fluorinated methyl isopropyl ether inhaled anesthetic, chemically related to desflurane and isoflurane. The incidence of abuse of sevoflurane is lower than that of other drugs of abuse possibly due to its inaccessibility to the general public and less pleasurable and addicting effects. The dead subject was an anesthetist found prone in bed holding an empty bottle of sevoflurane (Ultane). Serum, urine, and liver were screened for numerous drugs and metabolites using enzyme immunoassays and gas chromatography-mass spectrometry. Analysis did not reveal presence of any drug, including ethanol, other than sevoflurane. Sevoflurane was determined by headspace gas chromatography and revealed concentrations of 15 microg/mL in blood and 130 mg/kg in brain and lung. Autopsy revealed pulmonary edema and frothing in the lung, pathological findings associated with death by sevoflurane or hypoxia. The cause of death was ruled as sevoflurane toxicity and the manner of death as accident.

[Uptake and distribution of sevoflurane].

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Sevoflurane is characterized by a low blood/gas partition coefficient of 0.69, only desflurane and nitrous oxide have lower blood/gas solubilities. Alveolar equilibration is fast, a feature useful for rapid induction of anesthesia. Because of its pleasant smell, mask induction is feasible and routinely used in clinical settings. Formation of inorganic fluoride by metabolism and by compound A by degradation in CO₂-absorbent has not yet been shown to be nephrotoxic in humans. Pulmonal elimination of sevoflurane is rapid because of its low blood solubility. Clinical results showed that rapidity of recovery from sevoflurane anesthesia is equal to that of desflurane anesthesia. Physicochemical properties of sevoflurane allow its application in conventional vaporizers.

Desflurane anesthesia shifts the circadian rhythm phase depending on the time of day of anesthesia.

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Desflurane is one of the most frequently used inhalational anesthetics in clinical practice. A circadian rhythm phase-shift after general anesthesia with sevoflurane or isoflurane has been reported in mice, but few studies have reported this effect with desflurane. In the present study, we examined the rest/activity rhythm of mice by counting the number of running wheel rotations, and we found that desflurane anesthesia caused a phase shift in the circadian rhythm that was dependent on the time of day of anesthesia. We also found that desflurane anesthesia altered the relative mRNA expression of four major clock genes (Per2, Bmal, Clock, and Cry1) in the suprachiasmatic nucleus (SCN). These results are important for elucidating the effects of desflurane on the SCN, which is the master clock for the mammalian circadian rhythm. Further studies on the relationship between anesthesia and circadian rhythm may lead to the prevention and treatment of postoperative complications related to circadian rhythms.

The effects of isoflurane, sevoflurane, and desflurane anesthesia on neurocognitive outcome after cardiac surgery: a pilot study.

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Inhalation anesthetics such as isoflurane, sevoflurane, and desflurane are widely used in clinical practice; however, there is no study for comparing these drugs in cardiac surgery with respect to postoperative cognitive outcome and S100 beta protein (S100 BP) levels. In this study, we evaluated the effect of sevoflurane, isoflurane, and desflurane anesthesia on neuropsychological outcome and S100 BP levels in patients undergoing coronary artery bypass grafting (CABG) surgery with cardiopulmonary bypass (CPB). Forty-two male patients were prospectively randomized and classified into 3 groups according to the volatile agents used; isoflurane, sevoflurane, desflurane. All patients had a sufficient education level to participate in neuropsychological testing and a normal carotid Doppler ultrasonography. Blood samples for analysis of S100 BP were collected before anesthesia (T1), before heparinization (T2), 15 minutes into CPB (T3), following protamine administration (T4), postoperatively (T5), 24 hours after the operation (T6), postoperative day 3 (T7), and postoperative day 6 (T8). The neuropsychological tests, including Mini-Mental State Examination (MMSET) and visual-aural digit span test (VADST), were administered 1 day prior to surgery and on the third and sixth postoperative days. The postoperative third and sixth day MMSET scores and third day visual-written subtest scores in the sevoflurane group were significantly

lower than in the isoflurane and desflurane groups ($P < .05$). S100 BP levels increased with the beginning of anesthesia in the sevoflurane and desflurane groups. Although S100 BP decreased to baseline levels on postoperative day 1 in the sevoflurane group, this was significantly higher on the third and sixth days postoperatively in the desflurane group ($P < .05$). In the isoflurane group, the S100 BP level was significantly higher than the baseline level only after CPB ($P < .05$). Our study suggests that isoflurane is associated with better neurocognitive functions than desflurane or sevoflurane after on-pump CABG. Sevoflurane seems to be associated with the worst cognitive outcome as assessed by neuropsychologic tests, and prolonged brain injury as detected by high S100 BP levels was seen with desflurane.
